

1 Technical Project Description and Objectives

AI Agents for DAQ Services

The ATLAS collaboration faces evolving operational challenges heading into future data-taking periods, particularly concerning human resource optimization. In recent years, High Energy Physics (HEP) experiments have begun experiencing a shortage of available shifters and operators, a trend projected to intensify during Run 4 as personnel resources are drawn toward next-generation projects. While artificial intelligence (AI) methodologies, such as neural networks and deep learning, are well-established in offline data analysis, recent advancements in Generative AI and Large Language Models (LLMs) open new paradigms for accelerating automation within online Data Acquisition (DAQ) systems and experiment operations at large.

During standard operations, shifters frequently require expert support to resolve unexpected runtime anomalies or execute complex operational procedures. Although comprehensive documentation exists, it is often fragmented, poorly indexed, or structurally difficult to navigate under time-constrained shift conditions. Deploying an advanced, LLM-based AI-assistant to act as a natural language system entry point can mitigate these operational bottlenecks by providing rapid, intelligent access to system documentation and guiding operators, with different levels of expertise, through complex diagnostic workflows.

An effective and scalable implementation of this AI-assistant requires the deployment of specialized autonomous components, DAQ-agents, capable of programmatically interacting with and retrieving live status information from the core online services that form the backbone of the DAQ infrastructure. This framework will significantly streamline information access for both shifters and experts.

In this context, the Institute PROPER NAME HERE will contribute its expertise in artificial intelligence and software engineering to the design and development of these DAQ-agents. Specifically, the institute will develop modular agents to interface with the following core DAQ online services:

- **Run Control:** Granting the agent access to real-time status metrics of all applications within a data-taking session (e.g., Finite State Machine states, error and busy flags).
- **Process Management:** Providing access to low-level process diagnostics complementary to Run Control, including Process IDs (PIDs), file descriptors, and live CPU/memory consumption.
- **Configuration:** Allowing the agent to query the description of all DAQ elements within a data-taking session, alongside the parameters required for individual processes.
- **Information Service:** Interacting with the distributed data store where applications publish, update, and share structured runtime information.
- **Error Reporting System:** Retrieving messages, and their metadata, generated by applications and archived in a relational database.

In a first instance, the assigned tasks will focus on defining the interface protocols and prototyping individual agent modules. To optimize development and leverage existing infrastructure, this work package will directly integrate the knowledge and experience acquired within the CERN EP R&D Working Package 9.1. The final deliverables will mark a concrete milestone toward integrating production-ready AI assistants into ATLAS operations for Run 4.

The CERN team will retain ultimate responsibility for the overarching DAQ infrastructure and will set the operational directions and priorities for the integration.

Detector Control System

The ATLAS Detector Control System (DCS) monitors the sensors, power and safety systems of the ATLAS detector. It has recorded continuously since 2007, across all three LHC runs, spanning about 20 detector subsystems and more than 140 control systems that together monitor over 20 million parameters: currents, voltages, temperatures, cooling, gas and positions. A selected subset is written to the long-term archive, which today holds 68 TB in the ATLAS Oracle database and grows by roughly 9 TB per year. The archive is served through a modern time-series stack that answers multi-year queries in well under a second, complemented by natural-language access tooling already in use (a Model Context Protocol server over the archive and a documentation assistant).

In this context, the Institute of Engineering will contribute machine-learning developments on this archived data, in four work packages. All four use the archive as their data source, so students are productive remotely from day one, with no on-site access required. Each work package addresses a real operational need of the ATLAS DCS, matches the institute's expertise in anomaly detection on time series and logs and in big-data pipelines, and can lead to a technical publication.

DCS-WP1 - Early anomaly detection on detector sensor and power data. Train models to recognise when a channel drifts away from its normal behaviour before it triggers an alarm, and validate them against historical cases with known outcomes. Since normal behaviour drifts over eighteen years and labels are scarce, this is unsupervised, drift-aware anomaly detection on non-stationary multivariate series. It builds on a small anomaly-detection activity already running in the ATLAS Central DCS team, so students do not start from zero.

DCS-WP2 - Alarm-stream reduction and root-cause grouping. Apply log-clustering and grouping techniques to the DCS alarm stream: remove duplicates, group related alarms and surface the likely root cause, so that operators see ten real problems instead of a thousand lines. The challenge is turning bursty, high-volume alarm streams into a few ranked root causes - correlation and clustering, not just filtering.

DCS-WP3 - Shared data pipeline for machine learning. The common foundation: one clean, versioned, repeatable pipeline to extract and prepare archived DCS data for machine learning. It is developed jointly with the DAQ-side machine-learning and agentic-infrastructure activity described above, so that the collaboration builds one common stack rather than two parallel ones, and it is what keeps the work alive as student cohorts rotate.

DCS-WP4 - Predictive maintenance and hardware-failure prediction. Use the archived aging history of components (power supplies, cooling, front-end monitoring boards) to predict failures before they occur. Failures are rare and censored, so this is remaining-useful-life and survival analysis rather than simple classification. Directly useful for detector availability through the next long shutdown and the ATLAS upgrade.

Each work package is decomposed into six-month sub-projects matched to the institute's student cycle, each sized to a bachelor or master thesis while building toward the work-package goal. The first cycle starts with the most tractable sub-projects and the pipeline foundation; the harder anomaly-detection and failure-prediction work follows in later cohorts, reusing the foundation.

3.2 Top level work packages

#	Description
DAQ-WP1	Analysis of the available DAQ online services, including their exposed APIs and interoperability

DAQ- WP2	Prototype of an MCP framework model that could be adapted to all the DAQ online services
DAQ-WP3	Implementation and validation of a first agent interacting with one of the mentioned DAQ online services
DAQ-WP4	Implementation and validation of the remaining agents, reusing the defined MCP framework

#	Name
DCS-WP1	Early anomaly detection on sensor and power data
DCS-WP2	Alarm-stream reduction and root-cause grouping
DCS-WP3	Shared ML data pipeline (joint with DAQ activity)
DCS-WP4	Predictive maintenance and failure prediction

3.3 Deliverables

DAQ

- **Architecture:** A document detailing the agent-to-service communication protocols and the LLM orchestration layer.
- **DAQ-Agents:** Design and development of MCP (Model Context Protocol) agents enabling LLMs to interact directly with DAQ online services. Additionally, for each developed agent, documentation shall be provided summarizing its benchmarking results in terms of reliability, robustness, output consistency, performance and resource usage.

DCS

Every DCS deliverable comprises source code in the ATLAS GitLab repositories, with configuration, unit tests, continuous integration and deployment tooling, together with a technical note summarising the methods and benchmarking results (detection performance, robustness, false-positive rates, resource usage).

- 1 **Deliverable DCS-a - Shared feature-extraction pipeline, first release** (T0 + 6 months): reusable pipeline extracting and preparing archived DCS data for ML, shared with the DAQ-side activity.
- 2 **Deliverable DCS-b - Anomaly-detection prototype** (T0 + 12 months): models benchmarked against historical incidents on selected subsystems; production-quality release at T0 + 24 months.
- 3 **Deliverable DCS-c - Alarm-clustering prototype** (T0 + 18 months): duplicate removal, grouping and root-cause ranking evaluated on archived alarm streams.
- 4 **Deliverable DCS-d - Failure-prediction study and models** (T0 + 30 months): remaining-useful-life analysis on component aging histories, validated on known failures.
Deliverable DCS-e - Consolidated final release (T0 + 36 months): integrated stack, documentation and hand